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Investigating the Factors and Behaviors Influencing Entrepreneurial Intention in Agricultural Sectors among Indonesian Millennials

A F Suwanan [1] **and H R Allya**[2]
[1] Department of Economics, Universitas Negeri Malang, Indonesia.
[2] Department of Economics, Universitas Negeri Malang, Indonesia.

E-mail: ahmad.suwanan.fe@um.ac.id

Abstract. The most recent inventions and scientific breakthroughs are required since technological advancements cause such rapid development. To perceive prospects through entrepreneurship in the agriculture sector, it therefore takes a young generation that can make quick, technological innovations. Yet since 2014, there has been a decline in the workforce employed in agriculture. The purpose of this study is to look into the attitudes and circumstances that might encourage members of the younger generation to start their own businesses in the agricultural industry. The study was carried out in the following five Indonesian provinces: Banten, East Java, Central Java, and Yogyakarta Special Area. There were 153 respondents who participated in the research. The data has been analyzed using the Partial Least Square - Structural Equation Model (PLS-SEM) to understand the factor and behavior influencing agripreneurial intention among Indonesian Millenials. According to the research finding, there are several factors, namely perceived behavioral control and attitude towards behavior that have positive effect on the millennial's intention to become entrepreneurs in the agricultural sector. This shows that technical and soft skills have influenced the entrepreneurial incentive in the agriculture sectors. Yet, there is no relationship between the agricultural intention and subjective norms or outside motivation. Academic institutions, corporate leaders, and the government should take into account the demand for both soft and practical skills in the agricultural and industrial sectors for young people. Future labor force and added value will increase if younger generations participate more in farm sectors.

Keyword: agripreneurial intention, behaviour, PLS-SEM, subjective norm

1. Introduction

Agricultural sector has contributed significantly on decent work and economic growth. According to Food and Agriculture Organization (FAO), agriculture sector is the world's largest single employer, especially in developing countries where this sector has greater proportion in GDP. Agriculture is considered to be able to reduce poverty compared to other sectors, so that the agriculture has such an essential contribution on the Indonesian economy [1] [2] [3]. In the recent development, agriculture has developed rapidly and led to the use of more modern technologies [4] [5]. Technology-based agriculture or called smart agriculture has increased crop yields, minimize costs, and improve product quality through the use of modern systems [6] [7]. In Indonesian case, where the land ownership that is less than 0.2 ha/household, the smart agriculture is a must. Therefore, the younger generation who has capacity in developing technological progress through smart farming is considered capable of making rapid and modern changes to the agricultural sector.

In addressing the problem of agricultural sector, there are two approaches which have been discussed by previous researchers. First, the structural transformation model even with a more nuanced view of the role of agriculture still sees obstacles to industrialization. However, about 2.5 billion people still depend on this activity for their lives. Agriculture is considered to be able to reduce

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poverty [8] [9]. Second, agriculture for broader development, agriculture has changed drastically as a result of gobalization and technology. Globalization demands growth in the demand for agricultural exports, especially for products that have higher value and open up opportunities for developing countries to import food.

Indonesia is an agricultural country that has abundant natural resources, the agricultural sector in Indonesia has a great opportunity to be developed by young people. Currently, there is still a low contribution of the younger generation to the agricultural sector. Even though entrepreneurship in the agricultural sector is considered to be able to bring such a large profit. Therefore, a creative and innovative young generation is needed to be able to develop the agricultural innovation in ensuring the sustainability aspect and maintaining technological progress. During the Covid-19 pandemic, there have been progression in the development start-up industry in agriculture sector in Indonesia, for instance: iGrow, TaniHub etc. However the proportion of traditional farmers are remain dominant in total. Many of the traditional farmers have less entrepreneurial skill capacity. The lack of entrepreneurial skills and abilities are due to the discrepancies among farmers in human resource development. The dominant problems to become an entrepreneur in the agricultural sector are because of the financial reasons that require considerable financial resources [10]. People who have a background in agriculture areas tend to have interest to run firms in the non-agricultural areas [3].

The determinant of agricultur development has been discussed by many scholars. In the era of traditional agriculture, the lack of competence of farmers is due to the weak role of extension workers' agriculture. So that there is still a lack of efforts made to be able to improve the competence of farmers [11]. However, in the era of digitalization, farmers can find the right way through online learning center provided by government or private sectors. The issue that public should develop is regarding the intention to be entrepreneur for youth. The intention to be entrepreneurial in the agricultural sector is a desire that exists in a person to be able to build a business in agriculture. The intention possessed by a person to be entrepreneurial in agriculture means that they are interested in business in agriculture [12]. Currently, the opportunity to become entrepreneurial in the agricultural sector is very large, especially with the younger generation who can bring innovation to the agricultural sector, so as to provide sufficient employment for the younger generation [13].

This study aims to explain several factors and behaviors that can affect the intentions to be an entrepreneur for millennials in the agricultural areas. Understanding millenial's motivation in being an entrepreneur can lead better policy consideration to create better ecosystem for agribusiness sustainability. Motivation can be defined as encouragement, strength and driving force for someone to do something. This motivation usually arises due to several factors such as environment, education, experience in activities that have been participated in, age, access to information, land area, and facilities and infrastructure. This research is the primary study regarding the factor and motivation being entrepeneur in agricultur sector which conducted during the covid-19 Pandemic using respondents from 5 main provinces in Indonesia. This study has important implication in the ways stakeholders designing the agenda in improving entrepreneurial intention in agriculture sector, especially for millenials.

2. Materials and methods

This study is conducted using quantitative approach in explaining several factors and behaviors that can be the main driver for millenials in contributing agriculture sectors as an entrepreneur. This research uses primary data which collected from questionnaire using random sampling methord in five provinces, with a total of 153 respondents. The respondents in this study were students of universities from East Java, Central Java, West Java, Yogyakarta Special Region, and Banten which categorised as millenials. The definition of millenials in this study refers to a group of young people born in the early 1980s to early 2000s. this generation has comfortable with diversity, technology, and online communication to stay connected with friends.

This study examines the factors and behavior affecting entrepreneur intention in agriculture sector. According to the several studies, this research analyse the Attitude Toward the behaviour, Subjective

Norm, Perceived Behavioral Control as the independen latent. These variables were choosen due to the previous research conducted by some authors identifing the determinant of entrepreneurial intention. While the dependent variable is agriculture entrepreneurial intention this research calculates the data using Partial Least Square-Structural Equation Model (PLS-SEM) method. The range of likert scale are starting from 1 "strongly disagree" until 7 "strongly agree".
Research hypothesis:
H1: Attitude Toward the behaviour positively affects Agriculture Entrepreneur Intention
H2: Subjective Norm positivelty affects Agriculture Entrepreneur Intention
H3: Perceived Behavioral Control positively affects Agriculture Entrepreneur Intention

3. Results and discussion

There were 153 respondents who filled out the questionnaire. From the results of the questionnaire data that has been processed, overall respondents have an interest in entrepreneurship. However, those who are interested in the agricultural sector's entrepreneurship are not all and those who know about the agricultural sector are only a few. This could be reasonable due to the demography of respondents are come from wide range of study background including economics, sosiology, education and sciences.

3.1. Validity and Reliability Testing

According to the value of Average Variance Extracted (AVE) from Table 1, it can be seen that all latent variables have reached over 0.5. This means that the research variables could be declared to meet the validity aspect. The composite reliability from this study shows that all variables produce a Composite Reliability value of more than 0.7 so that it can be said to meet the reliability aspect.

Based on the results of data that has been processed using the PLS-SEM method, the attitude toward the behaviour variable has a positive effect on agriculture entrepreneur intention. So that the first hypothesis (H1) of this study is acceptable. Furthermore, the Subjective Norm variable negatively affects agriculture entrepreneur intention. The last variable perceived behavioral control has a positive relationship with agriculture entrepreneur intention. Therefore H1 and H3 are accepted. Meanwhile, H2 was declared rejected. The reason why only hypothesis 1 and hypothesis 3 are accepted will be described in the discussion section.

Table 1. Validity and Reliability Test

	Cronbach's Alpha	**rho_A**	**Composite Reliability**	**Average Variance Extracted (AVE)**
Attitude Toward the behaviour	**0.759**	**0.772**	**0.844**	**0.575**
Subjective Norm	**0.595**	**0.602**	**0.775**	**0.536**
Perceived Behavioral Control	**0.822**	**0.831**	**0.875**	**0.584**
Agriculture Entrepreneur Intention	**0.873**	**0.883**	**0.909**	**0.669**

Source : Processing Data from SmartPLS

Table 2. Hypotheses Testing

	Original Sample (O)	**Sample Mean (M)**	**Standard Deviation (STDEV)**	**T Statistics (|O/STDEV|)**	**P Values**	**Decision**
Attitude Toward the behaviour -> agriculture entrepreneur intention	0.273	0.274	0.057	4.757	**0.000**	H1 Accepted
Subjective Norm -> agriculture entrepreneur intention	0.031	0.037	0.050	0.628	**0.530**	H2 Rejected
Perceived Behavioral Control -> Agriculture Entrepreneur Intention	0.671	0.667	0.058	11.650	**0.000**	H3 Accepted

Source : Processing Data from SmartPLS

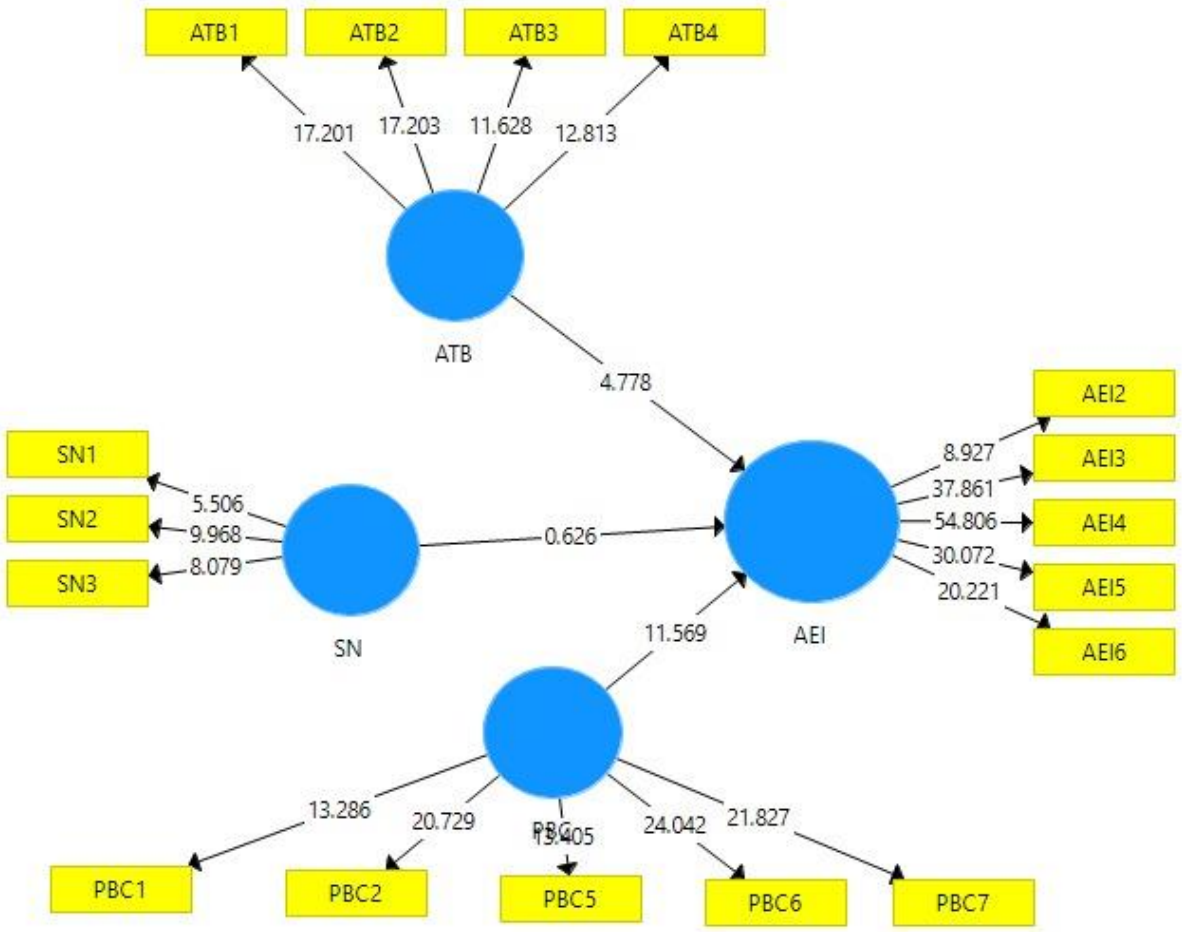


Figure 1. Path Analysis

3.2. Discussion

The results of this study show a significant correlation between Attitude Toward the behaviour (ATB) and Perceived Behavioral Control (PBC) on agriculture entrepreneur intention. However, there is no significant relationship between the Subjective Norm (SN) and the Agriculture Entrepreneur Intention (AEI). Agriculture Entrepreneur Intention is obtained from how high the interest in entrepreneurship in agriculture and interest in digital innovation in agribusiness in young people. Therefore the ATB has strong correlation to AEI as knowledge, network and ability to running the business is the important requirement in running the business entities.

Having the vision of agribusiness prospects in the future can become a substantial role for the younger generation to develop agriculture sector as an entrepreneurs. This is evidenced by the result of data collection that most of respondents are interested in promising prospects in the future in increasing employment opportunities. With the increase in employment, the agricultural sector is considered to be able to bring a bumper in economic development in Indonesia. Younger generation needs more opportunities and capacities in increasing ATB and PBC in the context of boosting the entrepreneurial motivation among millenials.

In the second tested variable, namely the subjective norm, it states that there is no influence between the background that a person has and the demands of any party to be able to be entrepreneurial in the agricultural sector. With the result of subjective norm calculation, it can be exhibited that the background and influence of parents, friends, teachers, and from several other parties cannot affect the intentions of young entrepreneurs in the agricultural sector. The conditions that occur here are referred to as external elements of the respondents, and cooperate positively with the intentions of the entrepreneur. In line with the results of research by Rizki et al [14] which states that subjective factors of norms or external factors cannot influence entrepreneurial intentions in the agricultural sector. However, the results of this research were contrary to the results of research [15] [16] [17]. which states that a person who has a background in agriculture and influence from external parties can influence the intention to become entrepreneurial in the agricultural sector.

Sufficient networks, knowledge of the agricultural sector, and financial adequacy can influence the younger generation to have a high interest in the desire for entrepreneurship in the agricultural sector. This indicates that ATB has the strong correlation with the entreprenurial intention. If millenials have sufficient capital, knowledge about agriculture, and their network, they are likely to start a business in the agricultural sector because it is a perceived ease to be able to open a business. Not in line with the results of Arisandi's research [18]. which states that the perception of control possessed by a person does not have a positive influence on entrepreneurial intentions in agriculture. In the research of Igwe et al [19] and Coviello [20] explained that a person who has a business in agriculture must develop a good network.

4. Conclussion

In understanding the factor and behavior among millenials to be an entrepreneur in agriculture sectors, perceived behavioral control and attitude towards behavior are the two main drivers for millennials to become entrepreneurs. This indicates that the soft skills and technical competencies have contributed the entrepreneurial motivation in agriculture sectors. However the subjective norms or external motivation has no correlation with the entrepreneurial intention in agriculture sectors. To increase the entreprenurial motivation, stakeholders should consider the need of soft skill and hard skill in agriculture industry development. If younger generation has higher participation in agriculture sectors, there will be higher labor force and greater economic added value in the future.

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References

[1] M. Ivanic dan W. Martin, “Sectoral Productivity Growth and Poverty Reduction: National and Global

Impacts," *World Dev.*, vol. 109, hal. 429–439, 2018, doi: 10.1016/j.worlddev.2017.07.004.
[2] L. Christiaensen dan W. Martin, "Agriculture, structural transformation and poverty reduction: Eight new insights," *World Dev.*, vol. 109, hal. 413–416, 2018, doi: 10.1016/j.worlddev.2018.05.027.
[3] J. Parcell, "Undergraduate Perceptions Of The Need For An Agricultural Entrepreneurship Curriuculum," no. January 2003, 2020.
[4] D. Byerlee, A. de Janvry, dan E. Sadoulet, "Agriculture for Development: Toward a New Paradigm," *Annu. Rev. Resour. Econ.*, vol. 1, no. 1, hal. 15–31, 2009, doi: 10.1146/annurev.resource.050708.144239.
[5] P. Andrade-Sanchez dan J. T. Heun, "Understanding Technical Terms and Acronyms Used in Precision Agriculture," *Univ. Arizona Coop. Ext. - AZ1534*, hal. 1–5, 2010, [Daring]. Tersedia pada: http://arizona.openrepository.com/arizona/bitstream/10150/146427/1/az1534-2010.pdf.
[6] G. Aceto, V. Persico, dan A. Pescapé, "A Survey on Information and Communication Technologies for Industry 4.0: State-of-the-Art, Taxonomies, Perspectives, and Challenges," *IEEE Commun. Surv. Tutorials*, vol. 21, no. 4, hal. 3467–3501, 2019, doi: 10.1109/COMST.2019.2938259.
[7] A. Gacar, H. Aktas, dan B. Ozdogan, "Digital agriculture practices in the context of agriculture 4.0," *Pressacademia*, vol. 4, no. 2, hal. 184–191, 2017, doi: 10.17261/pressacademia.2017.448.
[8] W. C. NEELY, "The Agricultural Fair," *Soil Sci.*, vol. 41, no. 2, hal. 160, 1936, doi: 10.1097/00010694-193602000-00020.
[9] L. D. Luc Christiaensen, *Agriculture and Poverty Reduction in Africa*, vol. 56, no. 8. 2007.
[10] G. O. Boateng, A. A. Boateng, dan F. J. van Lenthe, "Barriers To Youthful Entrepreneurship in," *Glob. J. Bus. Res.*, vol. 8, no. 3, hal. 109–120, 2014.
[11] A. A. Managanta, "The Role of Agricultural Extension in Increasing Competence and Income Rice Farmers," *Indones. J. Agric. Res.*, vol. 3, no. 2, hal. 77–88, 2020, doi: 10.32734/injar.v3i2.3963.
[12] C. N. Oktavina dan T. Sugiarti, "Motivasi Dan Minat Mahasiswa Prodi Agribisnis Utm Terhadap Pekerjaan Wirausaha," *Agriscience*, vol. 1, no. 1, hal. 308–323, 2020, doi: 10.21107/agriscience.v1i1.8009.
[13] L. K. Njeru, "Youth in Agriculture; Perceptions and Challenges for Enhanced Participation in Kajiado North Sub-County, Kenya," *Greener J. Agric. Sci.*, vol. 7, no. 8, hal. 203–209, 2017, doi: 10.15580/gjas.2017.8.100117141.
[14] R. N. Ridha, B. Burhanuddin, dan B. P. Wahyu, "Entrepreneurship intention in agricultural sector of young generation in Indonesia," *Asia Pacific J. Innov. Entrep.*, vol. 11, no. 1, hal. 76–89, 2017, doi: 10.1108/apjie-04-2017-022.
[15] N. Shiri, D. Mohammadi, dan S. M. Hosseini, "Entrepreneurial Intentions of Agricultural Students: Effects of Role Model, Social Support, Social Norms and Perceived Desirability: Scholars Research Library", Archives of Applied Science Research," *Arch. Appl. Sci. Res.*, vol. 4, no. 2, hal. 892–897, 2012.
[16] H. M. V. Kumar, N. B. Chauhan, D. D. Patel, dan J. B. Patel, "Predictive factors to avoid farming as a livelihood," *J. Econ. Struct.*, vol. 8, no. 1, 2019, doi: 10.1186/s40008-019-0141-7.
[17] R. M. S. Wijerathna, H. V. A. Wickramasuriya, dan B. Marambe, "Factors predicting the intention of academics of faculties of agriculture in the state universities in Sri Lanka to engage in outreach activities," *Trop. Agric. Res.*, vol. 26, no. 2, hal. 285, 2015, doi: 10.4038/tar.v26i2.8092.
[18] D. Arisandi, "INTENSI BERWIRAUSAHA MAHASISWA PASCASARJANA INSTITUR PERTANIAN BOGOR PADA BIDANG AGRIBISNIS," 2016.
[19] A. Igwe, A. Ogbo, E. Agbaeze, J. Abugu, C. Ezenwakwelu, dan H. Okwo, "Self-Efficacy and Subjective Norms as Moderators in the Networking Competence–Social Entrepreneurial Intentions Link," *SAGE Open*, vol. 10, no. 3, 2020, doi: 10.1177/2158244020934878.
[20] N. Coviello dan H. Munro, "Network relationships and the internationalisation process of small software firms," *Int. Bus. Rev.*, vol. 6, no. 4, hal. 361–386, 1997, doi: 10.1016/S0969-5931(97)00010-3.